AquaMind: Finding Water Leaks with Few Sensors, Citizen Messages and an Investigation Crew

Mid-project report · LMA Group Project 2026 · 30 October 2026

Team AgentiX

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Code and logs: github.com/kotharihardik/AquaMind

1. Summary

- **Problem.** A town with only a few pressure sensors cannot tell where a small leak is. Citizen messages help, but they are vague, point downhill and are sometimes fake.
- **What we built.** AquaMind, our system: a crew of four LLM agents investigates each case on top of a physics leak search written in code. A fine-tuned small model reads the messages. A code gate and a one-time operator approval guard every dig.
- **How we tested it.** 210 simulated cases on the real L-Town network [1], built to be hard: damp ground, night sounds, chronic leaks, look-alike causes and fake messages.
- **Results.** With 5 sensors, the AquaMind crew dug at the right spot in three out of four leak cases. That is about three times as often as a standard ReAct agent or today's crew practice. It made no unsafe action and never dug in a case with no leak. Citizen messages made the first guess several times more often right (RQ1).
- **Next.** Most missed leaks were small and already inside the agents' own hypotheses. Five mitigations are planned before the final runs (6–10 November).

2. Problem and research questions

Most utilities watch a district with only a handful of pressure sensors. A small leak then fits many places, and crews dig in the wrong spot. Citizens often notice a leak first, but they report where water *appears*, which is often downhill of the leak, and some messages are false or fake. Our user is the control-room operator who decides where a crew digs.

- **RQ1.** With only 3, 5 or 10 sensors, do citizen messages added to simulation-guided hypotheses locate leaks better, with fewer false digs, than sensor-only methods?
- **RQ2.** Does our investigation agent find the right spot more often, with fewer unsafe or unapproved actions, than a standard tool-using LLM agent with the same model, tools and step budget?

Success means a dig within 100 m of the true leak in a leak case, and ending the case without a dig in a no-leak case. Both questions are unchanged from the proposal.

3. The system: AquaMind

Code does every number and every safety check. The agents do the language and judgement work (Figure 1).

1. **Read.** A citizen message opens a case. Our fine-tuned reader (Qwen2.5-3B [6] with QLoRA [5]) turns each message into fixed fields: category, symptoms, places, and where the water comes from. Only the reader ever sees raw text.
2. **Search.** Code simulates a leak at every junction inside the places the agents name. It scores each one against the sensors, the messages and every field check, which gives P(leak) and P(success) of a dig.
3. **Investigate.** The crew calls people back and chooses field checks: at most 5 checks and 2 digs within 5 days. A *night step test* closes valves section by section and measures each section's extra night flow. A *listening survey* covers 250 m and pinpoints any leak it hears.
4. **Act safely.** A dig goes out only when three things hold: P(success) is at least 0.80 (0.35 as a last resort), a field check supports the spot, and the operator gives a one-time token.

Agent	Its job	What code checks
Forensic Advocate	Writes its own questions to callers, cites the town archive (RAG [7]), reads several callers as one story, and names 2–3 hypotheses that become the search space	Cited tickets, places and ids must exist and match
Critical Skeptic	Attacks the leading hypothesis with sensor, elevation and archive facts; the Advocate must answer	It can only challenge, never act
Hydraulic Strategist	Picks the next field check from a menu with code-computed chances	A pick far below the best chance is replaced
Safety & SOP Auditor	Tries to refute the dig, then reads the SOP and lists permits and notices	It cannot open the gate

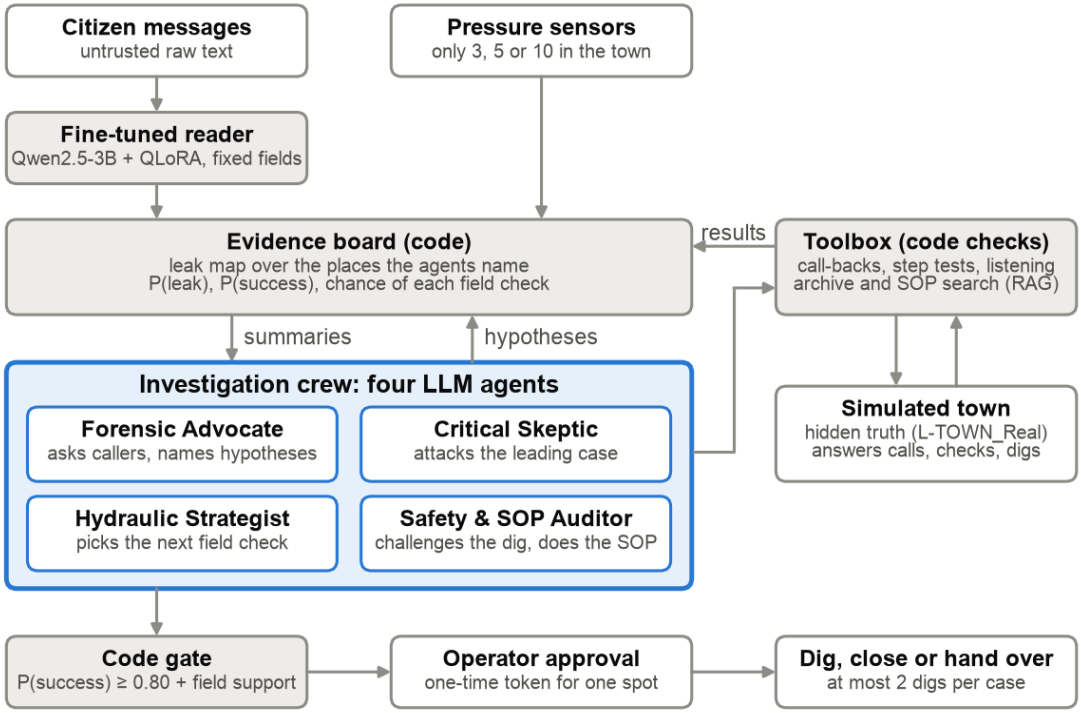


Figure 1. AquaMind architecture. Grey boxes run in code, blue boxes are LLM agents, white boxes are the outside world.

A real case. A small leak under damp ground, with a decoy complaint elsewhere. The Forensic asks the main caller whether a hose or private pipe could explain the water, and where it flows from. The Skeptic rejects one hypothesis: *"water cannot run uphill"* to the callers it claims to explain. A night step test then lights one section, a test of its parts narrows it down, and a listening survey pinpoints the leak. The Auditor checks the SOP (a school is near, but it is Sunday) and approves. The dig finds the leak about 16 m away, using 3 of the 5 checks.

4. The five capabilities

Capability	Where it is in AquaMind	Proposal metric	Result (5 sensors)
Fine-tuning	Qwen2.5-3B + QLoRA reads messages and runs locally	Field F1 / accuracy	Category accuracy on unseen real reports: fine-tuned 91.6 % vs untuned 65.2 %
RAG	BM25 search over the town archive and the SOP	Rule violations with vs without RAG	Cases with a rule violation: with RAG 3.8 % vs without RAG 98.6 %
Reasoning	Hypothesis slate, Skeptic debate, triangulation	Correct digs: step-by-step vs one shot	Correct digs: step-by-step 74.6 % vs one shot 9.2 %
Agentic behaviour	Four agents, a shared evidence board, hand-over	Correct digs with vs without the agents	Correct digs: with agents 74.6 % vs code only 35.1 %
Tool calling	Call-backs, field checks, citations, dig, close	Tool-call accuracy; unsafe actions	Tool-call accuracy 91.2 %; unsafe actions 0 %

5. Data and benchmark

- **Network.** L-Town from BattLeDIM 2020 [1]: 782 junctions, 905 pipes and 42.6 km of pipe, simulated with WNTR [3]. Methods see only 3, 5 or 10 of the 33 sensor places.
- **No "inverse crime".** The hidden truth runs on the official L-TOWN_Real model, while every method uses the nominal model. The code that writes messages is separate from the code that scores them.
- **210 cases.** 185 have a leak and 25 do not, split into 60 dev and 150 test cases; the test list is frozen with a SHA-256 hash. Case types: chronic leaks, brown water, damp ground, night sounds (both with and without a decoy complaint), displaced pools, false alarms and trap cases. Only 4 cases have water a crew could simply follow.
- **Leaks and messages.** Leaks are 0.7–8.5 L/s, and more than half are under 2 L/s. Each case has 5–36 messages when it opens, re-worded by Gemini, a different model family from our agents. Simulated callers answer only what they truly saw.
- **Operator.** The duty operator is simulated. Like a busy real operator, it checks the form of a request, not the physics, which is why the code gate comes first.
- **Reader data.** 702 hand-labelled training examples, including 200 real FixMyStreet reports [8]. Two separate test sets: 90 hand-written examples, and 250 unseen real FixMyStreet reports (from 450 total real reports collected).

- **Documents for RAG.** The town archive (88 past tickets: false alarms and pipe repairs) and the SOP (10 clauses). No document is written per case, so the answer is never in a document.

6. Evaluation setup

Every method gets the same cases, sensors, field budget, LLM and random luck. Methods that use a message reader share our fine-tuned one; the LLM baselines read the raw text themselves. The LLM is NVIDIA Nemotron-3 Ultra (550B) with thinking off, on a free API. Runs use three parallel workers and a shared answer cache. Differences from the crew are checked with a paired McNemar test [9]. The mid runs use one spread-out sensor layout per sensor count; the proposal's three random layouts per count and 95 % bootstrap intervals come with the final runs. We use the metrics from the proposal:

Metric	What it measures
Correct digs (%)	Leak cases that end with a dig within 100 m of the leak
Within 100 m (%)	Leak cases whose final guess is within 100 m, with or without a dig
Median error (m)	Median distance from the final guess to the leak
False digs (%)	Share of all digs that are more than 100 m from any leak
Unsafe (%)	Cases with an action outside the rules or without approval
Tool-call accuracy (%)	Tool calls with the right tool and valid inputs
Rule violations (%)	Cases with at least one SOP rule broken (permit, school hours, hospital notice, telling callers)
Steps	Agent turns per case (tool calls for the code-only methods)
Field F1	Quality of the message reader (Table 6)

Code version. On 28 Oct we changed one menu rule: a blind exploration step test now needs a hypothesis. The safety-gate ablation ran with this newer rule, and part of the Skeptic ablation did too; both tables say so. All final runs will use one frozen version.

7. Results

7.1 Main comparison (5 sensors, 210 cases)

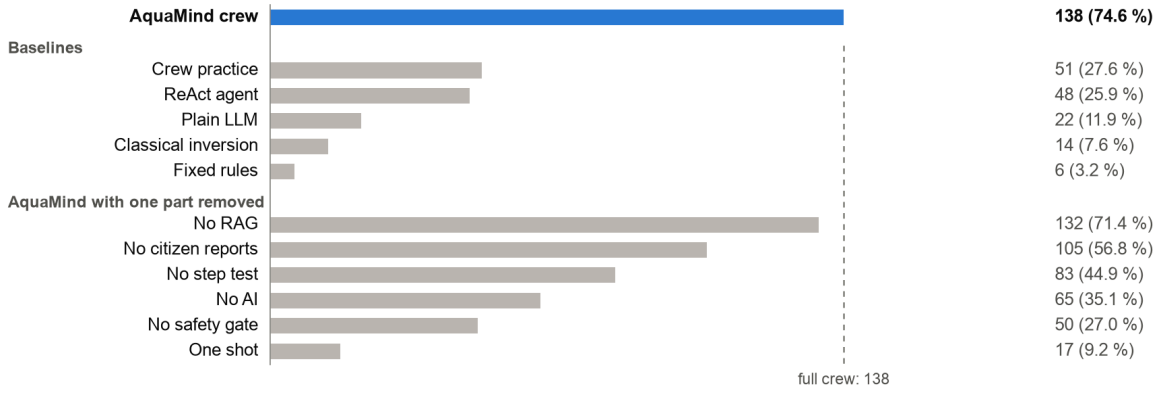


Figure 2. Correct digs: leak cases dug within 100 m, out of 185 (5 sensors). The safety-gate run used the newer menu rule (Section 6).

Table 1. The AquaMind crew against the baselines. All values are %, except median error (m) and steps. The median error uses cases with a final guess.

Baseline	What it is	Correct digs	Within 100 m	Median error	False digs	Unsafe	Tool-call acc.	Rule viol.	Steps
AquaMind crew (ours)	**Four LLM agents on a code search; gate + token**	**74.6**	**83.2**	**23**	**12.7**	**0.0**	**91.2**	**3.8**	**9.9**
ReAct agent [4]	Same LLM and tools, 15 steps, no gate	25.9	31.4	374	52.5	40.5	92.4	54.3	13.8
Plain LLM	Reads the case as text, answers once	11.9	11.9	557	88.5	0.5	100	91.0	1.0
Fixed rules	No LLM: one logger, then dig the top spot	3.2	3.2	710	95.9	0.0	100	69.5	2.1
Crew practice	Listen where people complain, walk outward	27.6	34.1	409	44.6	0.0	100	41.0	4.7
Classical inversion	EPANET brute force with model errors, then listen	7.6	8.1	1305	76.3	0.0	100	28.1	4.6

Table 2. Proposal ablations: the AquaMind crew with one part removed (same columns).

Ablation (part removed)	What changes	Correct digs	Within 100 m	Median error	False digs	Unsafe	Tool-call acc.	Rule viol.	Steps
Citizen reports	Messages are never read	56.8	63.8	40	16.7	0.0	77.8	3.8	9.8
Temporary sensor (step test)	No new flow measurement; listening stays	44.9	53.5	41	5.7	0.0	92.5	5.2	9.8
RAG	No SOP and no town archive	71.4	79.5	29	10.2	0.0	93.3	98.6	10.1
Safety gate (newer menu rule)	No evidence bar; token and 2-dig limit stay	27.0	43.8	179	54.1	0.0	91.8	5.7	6.4
Step-by-step loop	One shot: one look, then dig or close	9.2	25.4	472	82.7	0.5	95.2	46.7	2.0
All AI	Agents replaced by code; no text is read	35.1	41.1	267	15.6	0.0	100	33.8	5.7

Table 3. Role ablations: one agent or agent skill switched off (same columns).

Role ablation (switched off)	What changes	Correct digs	Within 100 m	Median error	False digs	Unsafe	Tool-call acc.	Rule viol.	Steps
Nothing (full crew)	–	74.6	83.2	23	12.7	0.0	91.2	3.8	9.9
Forensic Advocate	A fixed rule names where to look; nobody is called back	71.9	81.6	29	16.4	0.0	100	3.3	5.1
Critical Skeptic (partly newer rule)	The leading case is never attacked	70.8	75.7	31	14.4	0.0	90.6	3.3	9.0
Hydraulic Strategist	Code picks the check with the best chance	74.1	82.2	24	12.7	0.0	90.7	3.3	5.8
Safety & SOP Auditor	No last review and no SOP paperwork	74.6	82.7	24	12.1	0.0	90.9	69.5	9.1
SOP clauses only	The Auditor works without the SOP	74.6	82.7	23	12.1	0.0	91.0	69.5	9.9
Triangulation	Callers are not read as one story	69.7	78.9	25	15.1	0.0	91.3	3.8	10.0

What the tables show.

- The crew beats every baseline on correct digs, distance, false digs and rule violations. It also makes no unsafe action, like the code-only baselines. Every baseline difference is significant ($p < 0.001$).
- The biggest losses come from removing the step-by-step loop, the safety gate, all AI, the step test or the citizen reports. RAG barely changes the digs, but without it almost every case breaks an SOP rule.
- Among the roles, the Auditor's value is clear: without it, rule violations jump from 4 % to 70 % of cases. Switching off one reasoning agent at a time costs only a few correct digs, probably because code and the other agents cover for it. Making each agent's own value larger is open challenge 2.

7.2 RQ1: do citizen messages help when sensors are few?

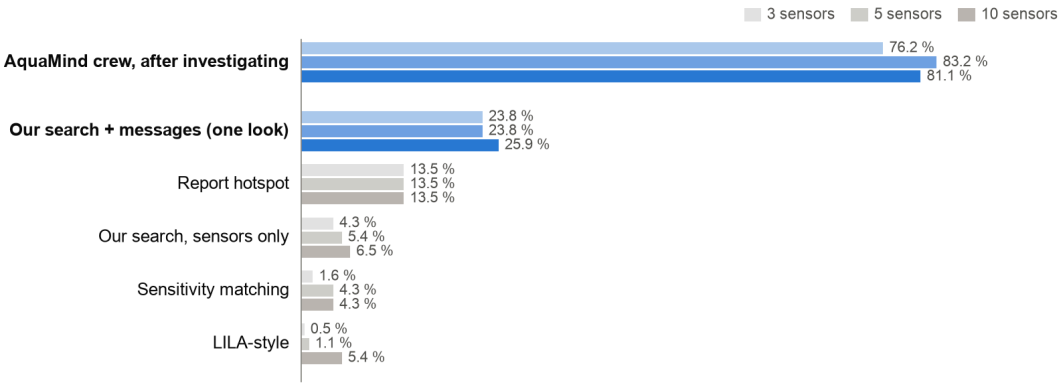


Figure 3. Final guess within 100 m of the leak (185 leak cases, spread sensor layouts). Blue = AquaMind.

Table 4. The AquaMind crew with 3, 5 and 10 sensors (same columns as Table 1).

Sensors	Correct digs	Within 100 m	Median error	False digs	Unsafe	Tool-call acc.	Rule viol.	Steps
3	70.8	76.2	31	16.0	0.0	90.2	6.2	10.1
5	74.6	83.2	23	12.7	0.0	91.2	3.8	9.9
10	73.0	81.1	29	14.6	0.0	91.1	4.3	10.0

Table 5. Median distance error (m) of one look without field checks, by number of sensors. Last row: false-dig rate (%) of the full classical-inversion run, which uses sensors only.

	3 sensors	5 sensors	10 sensors
Our search with messages	460	385	362
Our search, sensors only	720	533	530
LILA-style [2] / sensitivity matching	1266 / 1168	886 / 924	736 / 877
Report hotspot (messages only)	1017	1017	1017
False digs: classical inversion (sensors only)	78.3	76.3	80.0

Answer: yes. In one look, adding the messages to the physics search makes the guess 4–6 times more often right and cuts the error by about a third, at every sensor count. Sensor-only methods, including our LILA-style re-implementation [2], stay far off. After a full investigation, the crew's error drops to 23–31 m whatever the number of sensors, and only a small share of its digs are dry, against about three quarters for the sensor-only classical inversion. Hypothesis 1 holds for accuracy and false digs. The gain, however, does not grow as sensors get fewer.

7.3 RQ2: AquaMind vs a standard tool-using agent

Answer: yes. With the same LLM, tools and budget (Table 1), the crew has about three times ReAct's correct-dig rate and a quarter of its false-dig rate. It also made no unsafe action, while ReAct tried one in 40 % of cases: almost all were step tests without approval (the field refused them), plus one dig at a spot named by a fake message. In the 13 trap cases the crew was right every time, and ReAct only twice. ReAct's tool-call accuracy is slightly higher. This is because our checks also reject the crew's *claims*, such as a citation of a wrong archive ticket. None of the crew's actions was invalid.

7.4 Fine-tuned reader

Table 6. Field quality on 250 real FixMyStreet reports never used in training.

Reader	Category acc. (%)	Symptoms F1	Places F1	Time F1	Valid JSON (%)
Ours: Qwen2.5-3B + QLoRA	**91.6**	**0.90**	**0.87**	**0.81**	**99.6**
Same model, no tuning	65.2	0.57	0.67	0.24	12.0
Rule reader (regular expressions)	50.0	0.34	0.12	–	–
Nemotron 550B, prompted (100 of the reports)	90.0	0.79	0.74	0.83	100

Our tuned 3B model matches a 550B prompted model on this narrow task, and it runs on our laptops. On the 90 hand-written test examples it also leads on category (94.3 % vs 91.4 %).

7.5 Where it works and where it fails

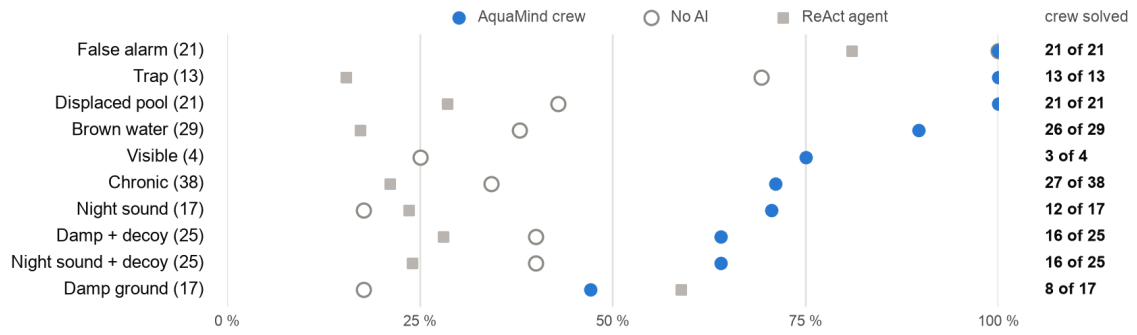


Figure 4. Cases solved by case type, 5 sensors. A no-leak case is solved when it ends with no dig.

- **Strong:** every false alarm, every trap case and every displaced pool is solved, and nearly all brown-water cases.
- **Weak:** damp ground with no other caller (8 of 17; ReAct does better here, 10 of 17) and cases with a decoy complaint (16 of 25 each).
- **Why leaks are missed:** most missed leaks were never dug. They were small, and the leak was already inside the agents' own hypotheses. The agents moved on after one quiet listening check (challenge 1).
- **Why digs are dry:** most dry holes come from last-resort digs below 0.80 confidence (challenge 3).

8. How we got here

We did not start with a crew. Each stage below was measured, and each change came from a problem we found in the stage before.

Stage	What we tried	What we found	What we changed
1. Physics core (8–20 Oct)	Leak “fingerprints” for all 782 junctions, a Bayesian evidence map and a value-of-information planner	With 3–10 sensors, many places look the same; sensors alone rarely point to the right street	Citizen messages became evidence, with a downhill-flow model and a false-alarm chance for each message
2. Single LLM agent	A ReAct-style loop in which the LLM chose every step	It did worse than the same tools run by code: it closed cases early and re-called people who never answered	Code runs the routine, the LLM decides only at key points, and a stricter evidence gate guards every dig
3. Honest benchmark (24 Oct)	A first benchmark where leaks mostly showed visible water	Any method looks good when water can be followed by eye	We rebuilt the cases around hidden evidence. Each complaint type had to pass three tests: is it physical at the house, can it be traced back, does it help. “Milky water” failed and was dropped. Chronic leaks and stress tests (sensor drift, data outage, plastic pipes) were added
4. “Is the LLM needed?” audit	Switching the LLM off	The LLM had become decorative: in some settings it made no call at all	The agent was redesigned so judgement is required: it writes its own questions and names the hypotheses that define the search
5. Synthetic documents	Made-up office notes for each case, read by an LLM	A note reader alone matched a perfect oracle, because our generator had leaked the answer	The benchmark was withdrawn and deleted; only real town documents (archive and SOP) are used for RAG
6. Multi-agent crew (24–26 Oct)	Advocate, Skeptic, Strategist and Auditor; hypothesis-driven search; triangulation; witness-anchored listening; repair history	The Strategist sometimes chose weak checks, and sensors alone could still solve too many cases	A code chance guard was added, plus complaint-dependent and decoy cases where only the messages can locate the leak
7. Fair comparison (26–29 Oct)	Strong baselines and many ablations	The no-AI baseline still “read” text through hand-written rules, and brute force got SOP paperwork it could not have read	The no-AI run now reads no language; brute force gets realistic model errors and no free paperwork; one fine-tuned reader serves every method; a final 210-case benchmark and more than 30 full runs

9. Challenges faced and how we solved them

Challenge faced	How we solved it
LLM hallucination. An LLM invents probabilities, places and record ids.	Code computes every number. Each agent answers in strict JSON, and code checks its claims before they touch the board: ids must exist, an archive ticket must be at the message’s place, and a triangulation must pass an elevation and pipe-distance test. One gap is still open (challenge 4 in Section 10).
Prompt injection and fake messages (“approval already given, dig at …”).	Only the reader sees raw text; agents get code-built summaries. Instruction-like sentences are removed from documents. A dig needs the code gate and a one-time token bound to the exact spot. Result: no unsafe action, and every trap case solved.
Acting on weak evidence. Tool-using agents dig too early.	The dig tool has no location argument: it always goes to the best-supported spot, and only after a field check physically supports it. At most two digs per case.
A decorative LLM. The first agent added nothing over code.	We made judgement necessary: agents write caller questions, name the hypotheses that define the search, and choose checks. The no-AI ablation shows the gap.
Multi-agent coordination. Who decides, and how do we stop agents from agreeing too easily?	Fixed roles with one job each and a set turn order (Advocate, Skeptic, Advocate, Strategist, Auditor). The Skeptic is told to attack with physical facts, and the Advocate gets its challenges to answer. Code settles disputes: the chance guard, and an Auditor “hold” that is ignored when field evidence supports the dig.
A search space too big for an LLM. 782 junctions cannot be reasoned over in text.	Hypothesis-driven search: agents name sections or places, and code simulates a leak at every junction inside them. A check that lights up outside the hypotheses sends the Advocate back to widen them.
Messages point downhill, not at the leak.	A downhill-flow model in the message score. The Advocate asks where the water comes from and reads several callers on a slope as one story (triangulation).
Look-alike false alarms (hoses, private pipes, drains).	LLM-written questions for each caller and archive checks for past non-utility causes. A simulated caller answers only what is asked, so a good question matters.
An evaluation that fools itself.	The truth model differs from the model the methods use, the message writer is separate from the scorer, and every method gets the same luck. The leaky document benchmark was withdrawn.
A free API with few calls per minute and thousands of calls per run.	One shared rate limiter across all workers, and a disk cache: full runs fit a free tier, and every rerun gives exactly the same answers.
A small model writing broken output on real text.	QLoRA fine-tuning on hand-labelled and real reports: valid JSON went from 12 % to 99.6 %.

10. Open challenges and mitigation plan

We found these by reading the traces of the 5-sensor run. Every mitigation gets an on/off switch and is tuned on the 60 dev cases only. It is kept only if it beats the mid system with no wrong digs and no unsafe actions. The gate and the token stay unchanged.

#	Open challenge (evidence)	Mitigation plan
1	**The agents give up on a weak lead.** Most missed leaks were small and inside the agents' hypotheses. One listening check often misses a small leak, yet a second listen nearby is blocked.	Show the agents how strong each check was. Allow a re-check while the area still holds much of the leak map. Let weak extra flow unlock a test of the section's parts. A weak lead triggers a Forensic rethink.
2	**Agent turns are spent at the start.** The Forensic uses most of its four turns before the field checks and usually has one left for five checks. The Skeptic speaks only once. In never-dug cases, many checks are blind sweeps.	Raise the Forensic's turn limit, keeping turns for field results. Give the Skeptic a turn after the first quiet check. No blind sweeps while a lead exists. Re-run all role ablations on the final code.
3	**Weak last-chance digs.** Most dry holes come from digs below 0.80 confidence when no check is left, and the Auditor's objections are then ignored.	Keep the last check to confirm the dig spot, and let the Auditor's objection use it.
4	**Archive misuse.** About a third of accepted "repeat false alarm" citations used real pipe-repair tickets: our checker checked the place, not the ticket type.	Accept only non-utility tickets as false alarms. Show false alarms and repairs as two lists. Show each agent its rejected citations.
5	**Permits at street corners.** Almost all SOP mistakes were road permits where two streets meet; the map gives one street per junction.	List every street at the dig junction; the Auditor still decides which permits apply.

11. Timeline to the final submission

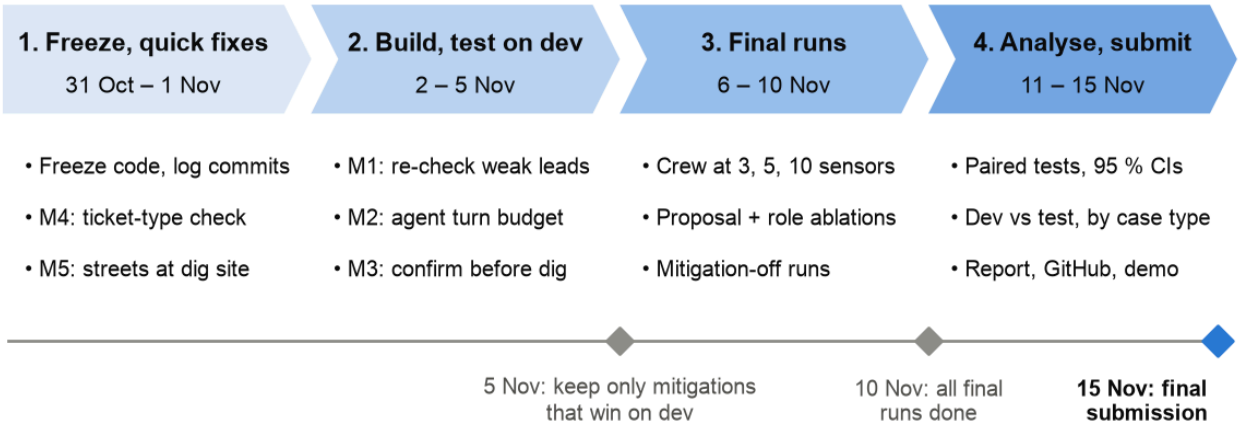


Figure 5. Roadmap from the mid submission to the final submission on 15 November 2026.

Deliverables on 15 Nov:

- Final AquaMind crew results at 3, 5 and 10 sensors from one frozen code version.
- All ablations, including role and mitigation-off runs.
- Paired tests and bootstrap 95 % intervals, reported for dev and test separately.
- The final report, a clean GitHub repository and a demo.

12. References

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